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Lecture due Mar 6, 2024 20:30 IST  
Recall from the course *Differential equations: 2 by 2 systems* that any second order linear ODE can be converted to a first order linear system called its **companion system**.

**Problem 10.1** Convert  $\ddot{x} + 5\dot{x} + 6x = 0$  to a first-order system of ODEs.

Solution

Define  $y := \dot{x}$ . Then

$$\begin{aligned}\dot{x} &= y \\ \dot{y} = \ddot{x} &= -5\dot{x} - 6x = -6x - 5y.\end{aligned}$$

In matrix form, this is  $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$  with  $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ -6 & -5 \end{pmatrix}$ . (The matrix  $\begin{pmatrix} 0 & 1 \\ -6 & -5 \end{pmatrix}$  arising this way is called the **companion matrix** of the polynomial  $x^2 + 5x + 6$ .)

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Conversely, given a first-order system, one can eliminate function variables to find a higher-order ODE satisfied by one of the functions. But usually we just leave it as a system.

**Problem 10.2** Given that

$$\begin{aligned}\dot{x} &= 2x - y \\ \dot{y} &= 5x + 7y,\end{aligned}$$

eliminate  $y$  to find a 2nd order ODE involving only  $x$ .

Solution

Solve for  $y$  in the first equation ( $y = 2x - \dot{x}$ ) and substitute into the second:

$$2\dot{x} - \ddot{x} = 5x + 7(2x - \dot{x}).$$

This simplifies to

$$\ddot{x} - 9\dot{x} + 19x = 0.$$

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Similarly, we can make companion systems for higher order linear differential equations by introducing new function variables for the derivatives.

Given an  $n^{\text{th}}$  order homogeneous linear ODE

$$D^n x + a_{n-1} (D^{n-1} x) + a_{n-2} (D^{n-2} x) + \cdots + a_1 (Dx) + a_0 x = 0, \quad \text{where } D = \frac{d}{dt},$$

define

$$x_0 = x \tag{4.62}$$

$$x_1 = Dx_0 = Dx \tag{4.63}$$

$$x_2 = Dx_1 = D^2 x \tag{4.64}$$

$$\vdots \tag{4.65}$$

$$x_{n-1} = Dx_{n-2} = D^{n-1} x. \tag{4.66}$$

Then the original order  $n$  DE can be rewritten as

$$Dx_{n-1} = -a_{n-1}x_{n-1} - a_{n-2}x_{n-2} - \cdots - a_1x_1 - a_0x_0.$$

Therefore, we have a system of  $n$  first order equations: the definitions of  $x_0, \cdots, x_{n-1}$ , and the original DE rewritten as above. In matrix form, the first order  $n \times n$  system is

$$\frac{d}{dt} \begin{pmatrix} x_0 \\ x_1 \\ \vdots \\ x_{n-1} \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{pmatrix} \begin{pmatrix} x_0 \\ x_1 \\ \vdots \\ x_{n-1} \end{pmatrix}.$$

$$\begin{pmatrix} x_{n-2} \\ x_{n-1} \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 & \cdots & 1 \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{n-1} \end{pmatrix} \begin{pmatrix} x_{n-2} \\ x_{n-1} \end{pmatrix}$$

The  $n \times n$  matrix on the right hand side is called the **companion matrix**, and this  $n \times n$  system is the **companion system** of the degree  $n$  ODE.

Note that the dimension of the companion system corresponds to the order of the single ODE we started with.

**Remark 10.3**     The same procedure of converting to a first order system works even if the starting ODE is inhomogeneous:

$$D^n x + a_{n-1} \left(D^{n-1} x\right) + a_{n-2} \left(D^{n-2} x\right) + \cdots + a_1 \left(Dx\right) + a_0 x = b.$$

The resulting companion system will have an extra vector on the right hand side:

$$\frac{d}{dt} \begin{pmatrix} x_0 \\ x_1 \\ \vdots \\ x_{n-2} \\ x_{n-1} \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{n-1} \end{pmatrix} \begin{pmatrix} x_0 \\ x_1 \\ \vdots \\ x_{n-2} \\ x_{n-1} \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ b \end{pmatrix}.$$

**Remark 10.4**     For constant coefficient ODEs, the characteristic polynomial of the higher order ODE (scaled, if necessary, to have leading coefficient **1**) equals the characteristic polynomial of the matrix of the first-order system.

More generally, we can also convert higher-order **systems** of ODEs to first order systems. For example, a system of **4** fifth order ODEs can be converted to a **20**  $\times$  **20** first order system ODEs. This is why it is enough to study first order systems.

### Companion system of third order ODE

1/1 point (graded)

Find the companion matrix **A** of

$$D^3 x + 4D^2 x + Dx - x = 0, \qquad \left(D = \frac{d}{dt}\right) \tag{4.67}$$

(Enter **[[a,b],[c,d]]** for the matrix  $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ . That is, use commas to separate entries within the same row and then enclose the entire row in **square brackets**. Then use **commas** to separate rows, and **square bracket** the entire matrix. Similarly for larger matrices.)

**A** = 

[[0,1,0],[0,0,1],[1,-1,-4]]

✔ Answer: [[0,1,0],[0,0,1],[1,-1,-4]]

**Solution:**

Given a third order linear ODE

$$D^3 x + 4D^2 x + Dx - x = 0, \qquad \left(D = \frac{d}{dt}\right)$$

Define

$$y = Dx, \tag{4.68}$$

$$z = Dy = D^2 x. \tag{4.69}$$

Then the DE can be written as

$$Dz = -4z - 1y + x.$$

Therefore, we have a system of three equations, which in vector notation is

$$\frac{d}{dt} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & -1 & -4 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}.$$

The **3**  $\times$  **3** matrix on the right hand side is the companion matrix.

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 Answers are displayed within the problem

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